

Real Estate Derivatives at Last? An Exploration of Prediction Markets

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Draft with data to 5 October 2026. Preliminary results.

Abstract

Exchanges have listed derivatives on property prices since 1991 and none has become liquid. Event contracts on prediction markets are the latest attempt. This paper asks whether they remove the obstacles that stopped earlier property derivatives, using all real estate contracts listed on Kalshi since June 2021 and the house price contracts on Polymarket. They do not. Kalshi's 1,669 real estate contracts account for 0.3 per cent of the volume in its economics category, and a real estate event trades about 71 per cent less than a comparable economics event. On Polymarket the median account traded 20 dollars and 90 per cent of accounts never returned. The contracts are quoted as tightly and priced as accurately as other economics contracts, so they work as forecasts, but they do not transfer risk. In 466,402 repeat sales from Cook County, Illinois, an index hedge removes at most 0.9 per cent of house-level price variance for holdings under two years, and no real estate contract on Kalshi runs longer than 18 months. The open order books would cover 6 per cent of a hedge for a single 500,000 dollar home.

AI disclosure: This paper was produced with extensive use of AI. Claude Code, an AI coding agent, searched for and downloaded the literature, collected the data from the public interfaces of the exchanges, wrote all of the code, ran the analysis, and drafted much of the text, tables and figures. I set the research question, directed each step, asked for the tests reported here, challenged results and writing that did not convince me, and decided what went into the paper. I did not write any of the code myself. The code is public, and the statements the text makes about the data are checked automatically against every new download. All errors are mine.

1 Introduction

Housing is the largest asset most households own, and they cannot hedge it. Real estate as a whole is the largest asset class without a liquid derivatives market (Fabozzi et al., 2020; Geltner and Fisher, 2007). Case et al. (1993) proposed cash-settled futures and options on house price indices to close that gap, and Shiller and Weiss (1999) proposed insurance built on them, but every attempt to list such contracts has failed for want of trade (Shiller, 2008). Thirty years on, Fabozzi et al. (2020) name four obstacles: settlement indices that fit the contracts badly, negligible liquidity in a market where nearly everyone wants the same side, the lack of an accepted pricing model, and regulation.

Prediction markets now list contracts on the same statistics. A contract pays one dollar if, say, a house price index ends above a stated level, and nothing otherwise. It differs from a futures contract in two ways that matter here. The payoff is a claim on a state, so that a ladder of strikes traces out a whole distribution, and the buyer is someone with a phone and a brokerage app, not an institution with a clearing account (Diercks et al., 2026). The hope is that speculators who enjoy taking a view on house prices will supply the liquidity that hedgers never supplied to each other, speculators being what makes a futures market deep enough for hedgers to use (Pagano, 1989). This paper tests that hope on Kalshi,¹ which Diercks et al. (2026) describe as the largest federally regulated prediction market, and on Polymarket.²

The contracts barely trade. All house price contracts listed on Kalshi since June 2021 traded 0.35 million one-dollar contracts (Table 1), whereas the CME housing futures that are usually described as a failure traded 612 million dollars of notional in their first eighteen months (Shiller, 2008), about 1,700 times as much. A real estate event trades roughly 71 per cent less than an economics event with the same listing period and number of strikes (Table 2). Polymarket’s public trade records show who takes part, which is mostly accounts that trade a few dollars once (Table 4).

I do not think this is a slow start that time will fix. The contracts are good forecasts, quoted on most days at the same spreads as other economics contracts and as well calibrated (Table 3), which is what the literature on prediction markets would lead one to expect (Wolfers and Zitzewitz, 2004). What they cannot do is hedge. Repeat sales in Cook County show that an index-settled contract removes almost none of an owner’s price risk at holding periods under two years (Figure 1), in line with Giacomelli (2021) and Eichholtz et al. (2021), and neither exchange lists anything longer. The order books, finally, are too thin for a single household (Table 5). The obstacles are the horizon, the basis and the missing counterparty, and neither a new payoff nor a better app does anything about those.

¹<https://kalshi.com>. Market data come from the exchange’s public API at <https://api.elections.kalshi.com/trade-api/v2>.

²<https://polymarket.com>. Event and trade data come from <https://gamma-api.polymarket.com> and <https://data-api.polymarket.com>.

2 Thirty years of property derivatives

The record is set out by Shiller (2008) and Fabozzi et al. (2020), and the facts in this section come from those two accounts unless another source is given. The Chicago Board of Trade considered a house price future in 1990 and dropped the idea after a survey in 1993 showed plenty of interest in protection against falling prices and little in selling it. The London Futures and Options Exchange launched property futures in May 1991 and terminated them in October, after little trade and a wash-trading scandal. Goldman Sachs listed covered warrants on UK house price indices in 2003, and a year later the open interest was roughly what would hedge a hundred houses. The CME launched futures and options on the S&P/Case-Shiller indices in May 2006; notional traded reached 612 million dollars by November 2007, open interest peaked at 109 million, and the market survives only in diminished form. Commercial property did somewhat better in the UK, where swaps on the IPD index traded more than three billion pounds of notional in the first quarter of 2007, than in the US, where no single index commanded the same acceptance (Geltner and Fisher, 2007).

Event contracts on housing are not new either. HedgeStreet offered consumers ten-dollar binary bets on house price indices for six cities in 2004 and closed for lack of trade, and UK spread-betting firms offered bets on house prices from 2001, which were seen as gambling and traded sparsely (Fabozzi et al., 2020; Shiller, 2008). Gürkaynak and Wolfers (2006) study auctions of digital options on macroeconomic data releases that began in 2002 and find the prices well calibrated, and Carlton (1984) mentions a proposal for futures on housing starts in the early 1980s. What is new is how many people trade event contracts.

The literature offers three main explanations for the thin markets. The first is the settlement index: repeat-sales indices are revised substantially and systematically downward as new sales arrive (Clapham et al., 2006), which is awkward for a contract that must settle on one published number. The second is basis risk, because a homeowner owns one house and not the index. de Jong et al. (2008) find that access to city-level housing futures improves a homeowner's welfare only slightly for that reason, Giacoletti (2021) shows that idiosyncratic risk accounts for a large fraction of house price variance, a fraction that falls as the holding period lengthens, and Eichholtz et al. (2021) show that for total returns the idiosyncratic part still makes up about half of risk at holding periods of fifteen years, a growing share of it coming from rental yields that are persistent at the level of the property. The third is the missing counterparty, since almost every household and lender is naturally long. Demand may also be weaker than the gains estimated by Englund et al. (2002) suggest: Sinai and Souleles (2005) point out that owning a home is itself a hedge against rent risk, and Caplin et al. (2003) report that take-up of subsidised home equity insurance in Illinois municipalities was 15 per cent at best.

3 Event contracts as claims on states

Let H_T be the value of a house price index published at date T . An event contract with strike K_j pays $\mathbf{1}(H_T > K_j)$. With strikes $K_1 < \dots < K_n$, the difference between neighbouring prices is the price of a claim on the interval,

$$q_j - q_{j+1} = \text{price of } \mathbf{1}(K_j < H_T \leq K_{j+1}), \quad (1)$$

and a homeowner who wants protection against a fall below K can approximate a put by buying the complements of the contracts with strikes below K , so that a dense ladder spans payoffs that a single futures contract does not (Breedon and Litzenberger, 1978; Ross, 1976). Under conditions set out by Wolfers and Zitzewitz (2006) the prices can be read as the market's probabilities, although Manski (2006) shows that with heterogeneous beliefs a price only bounds the mean belief of traders.

Each strike is a separate order book, however. Splitting trade across n strikes does not by itself make a hedge dearer, because a hedger who replicates a linear position over a given range of the index buys n contracts of one n -th the size, so that for a given spread the cost depends on the range covered and not on the number of steps. It costs more only if spreads widen or depth shrinks because each book receives a fraction of the order flow, which is what the model of Pagano (1989) predicts, in which depth depends on who else is expected to trade and trade tends to concentrate in one market. New contracts usually die in any case. Carlton (1984) finds that the majority of the futures markets listed in the United States between 1921 and 1983 failed within ten years, and among the features he names as necessary for success housing lacks one: prices that are highly correlated across the slightly different specifications and locations of the product.

The other obstacles stay where they were. The contract still settles on an index, the homeowner still owns one house, and the horizon moves in the wrong direction, because a household's exposure runs for years whereas event contracts resolve on the next data release. If the idiosyncratic share of house price variance is largest at short holding periods (Eichholtz et al., 2021; Giacomelli, 2021), an index contract hedges least at exactly the horizons that prediction markets list.

4 Data

The data are all markets in the economics category of Kalshi's public API, downloaded on 5 October 2026:³ 50,157 markets in 652 series and 4,588 events, the first of which opened in June 2021. A series is a recurring question, an event is one settlement date of that series, and a

³All code for this project is available at <https://github.com/thies/prediction-markets-real-estate>.

market is one strike within an event; volume is the number of one-dollar contracts traded over the life of the market.

I identify real estate series by hand from their titles and settlement sources and sort them into five groups: house prices (FHFA, Zillow and Canadian indices), rents (rent indices, CPI shelter, owners' equivalent rent), the 30-year mortgage rate, housing activity (sales, starts, permits, inventory), and credit and commercial real estate (mortgage and CRE delinquency, Manhattan office vacancy). Table 6 in the appendix lists every series. Everything else in the category forms the comparison group, which is dominated by Federal Reserve decisions, inflation, payrolls and petrol prices. Commercial property is nearly absent: Kalshi lists no contract on a commercial price, rent or return index, and the 3 series on data centre construction, data centre capacity and hotel room demand are very limited in scope and size, with 32 markets and 23,000 contracts between them, and stay in the comparison group.

The regressions use closed events, for which volume is final: 4,036 events in 432 series, of which 274 are real estate events (Table 8 in the appendix). For Polymarket I use the 64 house price events listed since January 2026, which settle on the Parcl Labs price index for the United States and 9 metropolitan areas, together with the record of every trade in them. Polymarket's other real estate contracts, 7 events on the mortgage rate since 2022 and 2 others, are too few to analyse.

5 Results

5.1 Volume

Table 1 gives the inventory. Real estate contracts account for 3.4 million of the 1,206 million contracts traded in the category, while the single series on Federal Reserve decisions traded 618 million. Economics is itself a small corner of the exchange. Of the 177 billion contracts traded on Kalshi in all categories, 73 per cent are bets on sporting events, about 38,000 times the volume of all real estate contracts together, and 640 individual sports series have each traded more than real estate as a whole (Table 7 in the appendix). House price contracts are spread over 28 series, most of which were listed for one event and not renewed (Table 6), and 56 per cent of rent markets never traded once. The longest real estate contract ran 537 days. Volume in the category is highly concentrated, ten series accounting for 81 per cent of it, so most of Kalshi's economics contracts are thin; real estate contracts are thinner still, the median market trading 284 contracts against 998 in the comparison group.

Table 1: **Kalshi economics contracts by group, June 2021 to October 2026**

	Series	Events	Markets	Contracts traded			Horizon (days)	
				Total (m)	Median	% zero	Median	Max
House prices	28	43	201	0.35	266	21	31	412
Rents	13	50	390	0.63	0	56	27	440
Mortgage rate	6	155	732	1.49	433	27	6	458
Housing activity	14	47	283	0.86	673	4	31	286
Credit and CRE	4	15	63	0.11	536	6	140	537
Other economics	587	4,278	48,488	1,202.28	998	20	1	4,175

Notes: One row per group of series. Median and share with zero volume are computed over markets (strikes). Horizon is the time from market open to close. Includes markets still open on the download date.

Table 2 regresses log event volume on a real estate indicator. Unconditionally a real estate event trades 81 per cent less than another economics event (column 1), and controlling for the listing horizon, the number of strikes and the quarter of closing narrows the gap only to 71 per cent (column 2). Volume rises with the listing horizon, as it must when volume is cumulative, and no differently for real estate (column 3); the more telling fact is that no long horizons are listed at all. Volume per strike falls with the number of strikes for real estate events, with an elasticity of about -0.6 , and not for the others (column 4). I do not read this as evidence of fragmentation, because only 23 real estate events list more than ten strikes, the exchange chooses the number of strikes with demand in mind, and quoted spreads do not widen with the number of strikes once controls are added (Figure 2 and Table 9 in the appendix).

Table 2: **Event volume: real estate against other economics events**

	(1)	(2)	(3)	(4)
	log volume	log volume	log volume	log volume per strike
Real estate	-1.637*** (0.374)	-1.247*** (0.442)	-0.906 (1.251)	-0.058 (0.441)
Real estate $\times$ log horizon			-0.129 (0.360)	
Real estate $\times$ log strikes				-0.788*** (0.282)
log horizon		0.516*** (0.116)	0.522*** (0.121)	0.479*** (0.104)
log strikes		1.120*** (0.203)	1.122*** (0.202)	0.155 (0.203)
Quarter fixed effects	No	Yes	Yes	Yes
Events	4,036	4,036	4,036	4,036
Series (clusters)	432	432	432	432
R^2	0.018	0.252	0.253	0.252

Notes: OLS on closed events. Volume enters as $\log(1 + \text{contracts traded})$. Standard errors clustered by series in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2 Quotes and price accuracy

Volume does not say what it would have cost to trade or whether the prices were any good. Daily closing quotes are available for every real estate market listed for at least two days and for a fixed sample of about a quarter of the other economics markets, which leaves 5,999 settled markets, 1,242 of them real estate (Tables 10 and 11 in the appendix). The median quoted spread is 6 cents in both groups, and with controls the two groups do not differ significantly (Table 3, column 1). Real estate markets have a two-sided quote on 70 per cent of days against 61 per cent elsewhere (column 2), which fits the account of Diercks et al. (2026), who note that Kalshi is supported by professional market makers. What is missing is anyone to take the other side: house price markets trade on 25 per cent of days and rent markets on 22 per cent, against 39 per cent in the comparison group. Nor is a spread of 6 cents cheap, since on a contract priced at 50 cents it amounts to a round trip of 12 per cent.

Prices are about as accurate as elsewhere. I take the quote midpoint one day before a market closes as its forecast of the probability that the contract pays out, and measure accuracy by the Brier score, the squared difference between that forecast and the outcome (one if the contract paid, zero if not), so that a perfect forecast scores zero and a forecast of 50 per cent always

scores 0.25. The raw score is worse for real estate, 0.120 against 0.091, but real estate prices sit closer to 50 cents, where any forecast scores worse, and conditional on $p(1 - p)$ the difference is 0.009 and insignificant, both one day and seven days out (columns 3 and 4). A regression of outcomes on prices gives a slope of 1.04 with no difference for real estate (column 5; Figure 3 in the appendix), which matches what Diercks et al. (2026) find for Kalshi's macroeconomic contracts. Of the four benefits that Fabozzi et al. (2020) list for property derivatives, information about future prices is thus the only one delivered, and even that rests on only 136 house price markets with a one-day forecast.

Table 3: **Quotes and forecast accuracy: real estate against other economics markets**

	(1)	(2)	(3)	(4)	(5)
			Brier score		
	Spread	Two-sided	1 day ahead	7 days ahead	Outcome
Real estate	-0.017 (0.015)	0.117*** (0.017)	0.009 (0.010)	0.006 (0.013)	0.016 (0.029)
Price one day before close					1.044*** (0.010)
Real estate $\times$ price					-0.017 (0.034)
$p(1 - p)$, one day			1.025*** (0.021)		
$p(1 - p)$, seven days				1.048*** (0.034)	
log horizon	-0.010*** (0.003)	0.055*** (0.006)	0.003 (0.002)	-0.002 (0.006)	
Quarter fixed effects	Yes	Yes	Yes	Yes	No
Markets	5,001	5,999	3,997	2,560	3,997
Events (clusters)	1,808	1,905	1,649	1,061	1,649
R^2	0.032	0.146	0.254	0.215	0.611

Notes: OLS on settled markets. Spread is the market's median quoted spread; two-sided is the share of days with both a bid and an ask. The Brier score in columns 3 and 4 is the squared difference between the outcome (1 if yes) and the price one day or seven days before close, lower being more accurate. Column 5 regresses the outcome on the price one day before close. Standard errors clustered by event in parentheses.

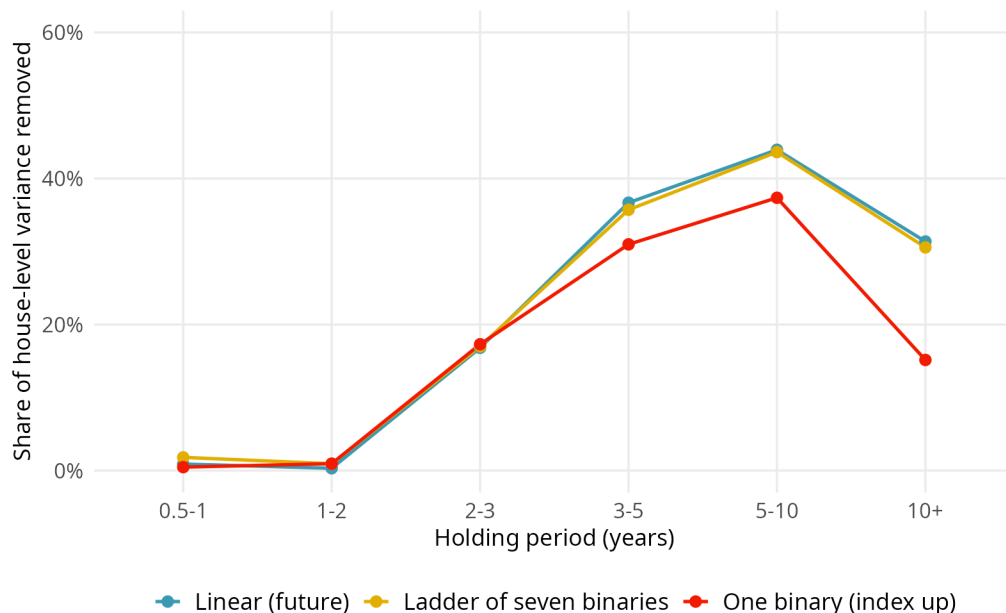
**** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

5.3 Hedging capacity of an index contract

Suppose the contracts were liquid. To see how much of an owner's price risk they could then remove, I use the Cook County Assessor's parcel sales (Cook County Assessor's Office, 2026). After dropping sales before 2000, multi-parcel sales, sales below 10,000 dollars, non-warranty deeds and flagged duplicates, consecutive sales of the same house or condominium at least six months apart give 466,402 repeat-sale pairs on 308,963 properties. For each pair, the log price change of the house is regressed on the payoff of a hedge written on the S&P/Case-Shiller Chicago index (S&P Dow Jones Indices, 2026) over the same months, and the R^2 is the share of house-level variance the hedge removes, the measure that Bertus et al. (2008) apply to Las Vegas.

Figure 1 gives the result by holding period (Table 12 in the appendix). For owners who sell within two years the index removes at most 0.9 per cent of the variance of their price change, and the share then rises to 17 per cent at two to three years, 37 per cent at three to five and 44 per cent at five to ten. This is the pattern that Giacomelli (2021) documents for California and Eichholtz et al. (2021) for Amsterdam, and it is the opposite of what the exchanges supply, the longest house price contract on Polymarket running 102 days. The payoff shape matters much less than the horizon: a ladder of seven binary contracts, the median number of strikes per house price event on Kalshi, removes as much variance as a linear position at every holding period, as the spanning argument of Ross (1976) implies.

Figure 1: **Hedgeable share of house price variance by holding period**



Notes: R^2 from Table 12.

The larger shares come from one episode, however. For houses bought in 2000–2011, whose

holding periods span the boom and the bust, the index removes 44 per cent of variance at three to five years, but for houses bought from 2012 on only 4 per cent, and in New York City, where the public sales file (New York City Department of Finance, 2026) starts in 2016, at most 0.2 per cent at any holding period (Tables 13 and 14 in the appendix). An index hedge protects against a market-wide crash and does little for the owner of one house in ordinary years. The shares are also upper bounds, since the hedge matches the holding period exactly, hedge ratios are fitted in sample and the index is taken as final, and they concern price risk only, whereas Eichholtz et al. (2021) show that rental yields account for a growing part of property-level risk at longer horizons. Short holding periods are a selected sample, heavy with renovated resales and distressed sales, so the result below two years should be read as no evidence of hedgeable risk and not as a precise zero.

5.4 Who trades

Polymarket’s 64 house price events traded 1.56 million contracts up to October 2026, for which takers paid 0.85 million dollars. That is more than Kalshi’s house price contracts and still a rounding error against the CME figure. The first monthly events, opened in January 2026, traded a median of 21,000 contracts, those opened from March to June 6,000, and the quarterly contracts that replaced them in July 100,000 (Table 15 in the appendix). I cannot say how much of the July jump is organic, because the exchange pays liquidity rewards on open contracts and does not report whether closed ones carried them.

Because Polymarket settles on a public blockchain, each trade carries the wallet addresses of both sides. A wallet is an account and not a person, but the addresses show what kind of participation there is. The house price markets saw 37,057 transactions among 3,443 wallets. The median wallet traded 20 dollars in total, 75 per cent traded less than 100 dollars, and 90 per cent appear in the contracts of a single launch month and never again. Table 4 sorts wallets by the share of their dollars that came from resting orders and by the ratio of their net position to their gross trading: 990 wallets with 34 per cent of the dollars provide liquidity and 1,291 wallets with 39 per cent take a position and hold it. The 19 wallets above 10,000 dollars are generalists. The median one has traded about 4,900 markets across the exchange and 12 of them have collected more than 1,000 dollars in liquidity rewards (Table 16 in the appendix), so that for them a house price contract is one more market to quote or one more bet. The largest traded 109,000 dollars. These are not the pension funds, insurers and hedge funds that Fabozzi et al. (2020) expected to carry the other side of a property derivatives market.

Table 4: **Wallets in Polymarket house price markets, by trading behaviour**

	Wallets	Dollars (%)	Median dollars	Above \$10,000
Takes and holds	1,288	39	5	9
Takes and trades out	701	23	20	5
Rests orders, ends flat	982	34	64	5
Rests orders and holds	461	4	12	0

Notes: All trades in Polymarket contracts on median home values, January 2026 to October 2026. Dollars are summed over both sides of each trade. A wallet rests orders if at least half its dollars came from orders filled by another party. It ends flat, or trades out, if the absolute net position summed over markets is less than half its gross trading, in shares.

5.5 A household's hedge

There is little sign that households use these contracts to hedge. A household hedging its home would trade the contracts for one area, come back each time they are relisted, keep what it buys, and stay on one side of the market's consensus. Of the 95 wallets that traded at least 1,000 dollars in two or more launch months, 3 concentrate four fifths of their trading on one area, of which 1 also kept a one-sided position, the largest having traded about 1,200 dollars.

Nor could the markets absorb a real hedge. An owner of a 500,000 dollar home who wants cover against a 10 per cent fall needs contracts that pay 50,000 dollars. Table 5 summarises the order books of all open house price contracts on 5 October 2026. Adding up, across the strikes of an event, the contracts on offer within five cents of the best quote gives a median of 3,300 dollars of payout per event on Kalshi and 2,600 on Polymarket, or 6 per cent of what this one household needs, and the deepest event offers 23 per cent. Prices would move long before that: an order for 100 contracts fills within five cents on 83 per cent of book sides, an order for 1,000 on 13 per cent, and the 7,000 contracts per strike that a seven-strike ladder would take on 1 per cent (Table 17 in the appendix). A single homeowner would exhaust the quotes of a metropolitan market, and would have to do so again every few months as contracts expire. The snapshot is of one day, and market makers may refill their quotes, but there is no sign of standing capacity on the scale of even a handful of houses.

Table 5: **Order book depth of open house price contracts**

	Events	Markets	On offer per event (dollars)			Whole book	
			At best	Near, median	Near, max	Median	Max
Kalshi	15	34	3,171	3,262	11,748	5,980	44,969
Polymarket	10	70	773	2,624	7,462	12,147	27,737

Notes: Order books of open contracts on house prices, recorded on 5 October 2026. Each contract pays one dollar, so the number of contracts on offer is the payout a trader could buy. Contracts are summed over the strikes of an event, separately for buyers of yes and buyers of no, and the deeper side is reported. At best is the median across events of the payout on offer at the best quote; near counts offers within five cents of it. The whole book includes offers far from the best quote.

6 Conclusion

Prediction markets have not given real estate the derivatives market it lacks. Event contracts on house prices, rents and mortgage rates are listed, quoted and accurately priced, and almost nobody trades them. They make up 0.3 per cent of the volume in Kalshi’s economics category, the accounts that trade them on Polymarket are small, do not return and include no one who looks like a hedger, and the open order books could not carry the hedge of a single home. What the contracts supply is a forecast of the next index release, which is useful and is not risk transfer.

The reasons are the old ones. Fabozzi et al. (2020) put liquidity and the missing counterparty at the centre, and the retail speculators who were supposed to fill that gap have come to the exchanges and trade sport, which makes up 73 per cent of Kalshi’s volume, and within economics the Federal Reserve and the price of petrol. Had they come to housing, the contracts would still not hedge much, because at the horizons listed an index removes almost none of the price risk of an individual house, and in calm markets it removes little at any horizon. A market that completes real estate would need contracts that run for years, on indices local enough to track a house, with counterparties willing to hold the other side for that long. Nothing in the event contract format rules that out, since a ladder of binary contracts hedges as well as a future, but nothing in the present markets points towards it either.

Several limits should be kept in mind. The hedging test uses Case-Shiller indices and not the Zillow and Parcel indices on which the event contracts settle, covers two cities, and fits hedge ratios in sample. Order book depth is observed on one day, since neither exchange archives its books, although the code behind this paper records a snapshot at every update, so that a history will build up. Polymarket’s quotes and price accuracy have not been analysed, and for the CME contracts after 2007 I rely on Fabozzi et al. (2020). These markets are young and changing, and the analysis is built to be rerun as they do.

References

- Mark Bertus, Harris Hollans, and Steve Swidler. Hedging house price risk with CME futures contracts: The case of Las Vegas residential real estate. *Journal of Real Estate Finance and Economics*, 37(3):265–279, 2008.
- Douglas T. Breeden and Robert H. Litzenberger. Prices of state-contingent claims implicit in option prices. *Journal of Business*, 51(4):621–651, 1978.
- Andrew Caplin, William Goetzmann, Eric Hangen, Barry Nalebuff, Elisabeth Prentice, John Rodkin, Matthew Spiegel, and Tom Skinner. Home equity insurance: A pilot project. Working Paper 03-12, Yale International Center for Finance, 2003.
- Dennis W. Carlton. Futures markets: Their purpose, their history, their growth, their successes and failures. *Journal of Futures Markets*, 4(3):237–271, 1984.
- Karl E. Case, Robert J. Shiller, and Allan N. Weiss. Index-based futures and options markets in real estate. *Journal of Portfolio Management*, 19(2):83–92, 1993.
- Eric Clapham, Peter Englund, John M. Quigley, and Christian L. Redfearn. Revisiting the past and settling the score: Index revision for house price derivatives. *Real Estate Economics*, 34(2):275–302, 2006.
- Cook County Assessor’s Office. Assessor – parcel sales. Cook County Open Data, dataset wvhk-k5uv, <https://datacatalog.cookcountyi1.gov/d/wvhk-k5uv>, 2026. Accessed 4 October 2026.
- Frank de Jong, Joost Driessen, and Otto Van Hemert. Hedging house price risk: Portfolio choice with housing futures. Working paper, SSRN 740364, 2008.
- Anthony M. Diercks, Jared Dean Katz, and Jonathan H. Wright. Kalshi and the rise of macro markets. Finance and Economics Discussion Series 2026-010, Board of Governors of the Federal Reserve System, 2026.
- Piet Eichholtz, Matthijs Korevaar, Thies Lindenthal, and Ronan Talleg. The total return and risk to residential real estate. *Review of Financial Studies*, 34(8):3608–3646, 2021.
- Peter Englund, Min Hwang, and John M. Quigley. Hedging housing risk. *Journal of Real Estate Finance and Economics*, 24(1/2):167–200, 2002.
- Frank J. Fabozzi, Robert J. Shiller, and Radu S. Tunaru. A 30-year perspective on property derivatives: What can be done to tame property price risk? *Journal of Economic Perspectives*, 34(4):121–145, 2020.

- David Geltner and Jeffrey D. Fisher. Pricing and index considerations in commercial real estate derivatives. *Journal of Portfolio Management*, 33(5):99–118, 2007.
- Marco Giacoletti. Idiosyncratic risk in housing markets. *Review of Financial Studies*, 34(8):3695–3741, 2021.
- Refet Gürkaynak and Justin Wolfers. Macroeconomic derivatives: An initial analysis of market-based macro forecasts, uncertainty, and risk. Working Paper 11929, National Bureau of Economic Research, 2006.
- Charles F. Manski. Interpreting the predictions of prediction markets. *Economics Letters*, 91(3):425–429, 2006.
- New York City Department of Finance. NYC citywide annualized calendar sales update. NYC Open Data, dataset w2pb-icbu, <https://data.cityofnewyork.us/d/w2pb-icbu>, 2026. Accessed 4 October 2026.
- Marco Pagano. Trading volume and asset liquidity. *Quarterly Journal of Economics*, 104(2):255–274, 1989.
- Stephen A. Ross. Options and efficiency. *Quarterly Journal of Economics*, 90(1):75–89, 1976.
- Robert J. Shiller. Derivatives markets for home prices. Working Paper 13962, National Bureau of Economic Research, 2008.
- Robert J. Shiller and Allan N. Weiss. Home equity insurance. *Journal of Real Estate Finance and Economics*, 19(1):21–47, 1999.
- Todd Sinai and Nicholas S. Souleles. Owner-occupied housing as a hedge against rent risk. *Quarterly Journal of Economics*, 120(2):763–789, 2005.
- S&P Dow Jones Indices. S&P CoreLogic Case-Shiller home price indices for Chicago and New York, not seasonally adjusted. Series CHXRNSA and NYXRNSA, retrieved from FRED, Federal Reserve Bank of St. Louis, <https://fred.stlouisfed.org/series/CHXRNSA>, 2026. Accessed 5 October 2026.
- Justin Wolfers and Eric Zitzewitz. Prediction markets. *Journal of Economic Perspectives*, 18(2):107–126, 2004.
- Justin Wolfers and Eric Zitzewitz. Interpreting prediction market prices as probabilities. Working Paper 12200, National Bureau of Economic Research, 2006.

A Supporting tables and figures

Table 6: Real estate series on Kalshi

Series	Title	Events	Markets	Contracts (000)	First	Last
<i>House prices</i>						
KXHOMEUS	US housing price up monthly	6	26	75.1	2022-07	2023-06
KXHOMEUSY	US housing prices up	3	10	48.8	2023-02	2025-12
KXEXUSHOME	Median Existing US Home Price	3	27	39.6	2026-07	2026-10
KXCAHPIYOY	California home prices	1	1	20.5	2026-08	2027-02
KXTORCONDO	Toronto condo price below \$X in [year]?	1	1	18.2	2026-05	2027-02
KXILHPIYOY	Illinois home prices	1	1	14.1	2026-08	2027-02
KXNVHOMEPRICE	Nevada House Price Index	1	1	12.5	2026-08	2027-02
KXNHHOMEPRICE	New Hampshire House Price Index	1	1	12.3	2026-08	2027-02
KXHPI	Housing Price Index	3	3	11.9	2021-12	2022-02
KXUSHOME	Median US New Home Price	4	48	11.2	2026-07	2026-10
KXEDMHOME	Edmonton home price below x in [year]	2	2	10.1	2026-06	2027-01
KXMTLHOME	Montreal single-family home price below X in [year]?	1	1	9.7	2026-06	2027-01
KXOTTAWAHOME	Ottawa home price below X in [year]?	1	1	9.5	2026-06	2026-08
KXCOHPIYOY	Colorado home prices	1	1	9.4	2026-08	2027-02
KXBOSHOMIEVAL	Boston typical home value for date	1	8	8.4	2026-03	2026-04
KXUSHOMIEVAL	US Metro typical home value for date	1	8	7.7	2026-03	2026-04
KXCALHOME	Calgary home price below X in [year]	1	1	5.2	2026-06	2027-01
KXRIHOMEPRICE	Rhode Island House Price Index	1	1	5.2	2026-08	2027-02
KXHOUHOMIEVAL	Houston typical home value for date	1	8	3.8	2026-03	2026-04
KXSEAHOMIEVAL	Seattle typical home value for date	1	8	3.6	2026-03	2026-04
KXSFHOMIEVAL	San Francisco typical home value on date	1	8	3.1	2026-03	2026-03
KXDCHOMIEVAL	Washington DC typical home value for date	1	8	3.0	2026-03	2026-03
KXCANHOME	Will Canada's national average home price exceed X i	1	1	2.2	2026-05	2027-01

Series	Title	Events	Markets	Contracts (000)	First	Last
KXVANCONDO	Will the average Vancouver condo price fall below X	1	1	2.2	2026-06	2027-01
KXMIAHOMEVAL	Miami typical home value for date	1	8	1.3	2026-03	2026-03
KXDENHOMEVAL	Denver typical home value for date	1	8	1.3	2026-03	2026-04
KXPHXHOMEVAL	Phoenix typical home value for date	1	8	1.0	2026-03	2026-03
KXUTHOMEPRICE	Utah House Price Index	1	1	0.8	2026-08	2027-02
<i>Rents</i>						
KXNYCASKRENT	NYC median asking rent	1	8	169.1	2026-08	2027-01
KXNYCRENTSY	NYC rent yearly increase	2	16	156.4	2025-07	2027-01
KXCPISHELTER	CPI on shelter	18	90	119.9	2023-12	2025-06
KXSFRENTSY	SF yearly rent growth	2	7	53.3	2024-04	2026-01
KXMEDIANRENTLA	MEDIAN RENT LA	1	5	45.5	2025-11	2025-12
KXOER	Owners' equivalent rent for [month]	2	40	24.0	2026-08	2026-10
KXMEDIANRENTCHI	MEDIAN RENT CHICAGO	1	5	15.9	2025-11	2025-12
KXMEDIANRENTMIA	MEDIAN RENT MIAMI	1	5	13.2	2025-11	2025-12
KXRENTNYCY	Rent price increase in NYC yearly	3	10	13.2	2023-02	2025-12
KXRENTNYCM	Rent price increase in NYC monthly	6	23	9.1	2022-07	2023-07
KXMEDIANRENTHOU	MEDIAN RENT HOUSTON	1	5	9.0	2025-11	2025-12
KXMANHATTANRENT	The average Manhattan rent in [month]	1	11	1.4	2026-08	2026-09
KXTRUFHOU CPI	Truflation US CPI Housing Inflation Index	11	165	0.3	2026-06	2026-07
<i>Mortgage rate</i>						
KXFRM	Mortgage 30-yr rate this week	132	505	720.8	2021-07	2024-02
KXFRMMAX	Mortgage rate high	6	24	330.8	2022-05	2024-12
KX30YMORTW	30-year mortgage rate this week	7	124	198.6	2026-08	2026-10
KXMORTGAGERATE	Mortgage Rate	2	32	147.1	2026-01	2027-12
KXFM30YMTG	Freddie Mac 30Y mortgage rate	3	27	48.6	2026-04	2027-12
KXFRMMIN	Mortgage rate low	5	20	43.7	2023-10	2024-12
<i>Housing activity</i>						
KXHOUSINGSTART	Monthly housing starts	7	58	203.1	2026-03	2026-10
KXCTMFPERMITS	Connecticut multifamily housing permits	1	8	123.7	2026-08	2027-05

Series	Title	Events	Markets	Contracts (000)	First	Last
KXVTPERMIT	Vermont housing units au- thorized	1	11	115.0	2026-08	2027-05
KXCOBUILDPERMIT	Colorado housing units au- thorized by building permit	1	5	78.2	2026-08	2027-01
KXNHSALES	New home sales in time pe- riod	7	56	69.3	2026-03	2026-10
KXBUILDPERM	Building permits for month	7	51	63.0	2026-03	2026-10
KXEHSALES	Existing home sales in time period	7	56	58.2	2026-03	2026-10
KXBOZPERMIT	Bozeman housing units au- thorized	1	6	47.2	2026-08	2027-05
KXHOME	New home sales	7	7	47.1	2021-06	2022-03
KXUSHOMEINVENT	US Active Housing Inventory	2	14	23.5	2026-09	2026-11
KXCANHOUSTART	Canada housing starts above X in [year]?	2	2	20.7	2026-05	2027-01
KXCABUILDPERMIT	California permitted housing units	1	6	12.8	2026-08	2027-01
KXHOUSELENGTH	US house average sale length	2	2	0.4	2023-08	2023-10
KXHOUSESTART	US housing starts rise	1	1	0.1	2023-09	2023-09
<i>Credit and CRE</i>						
KXCREDEF	Commercial real estate de- faults	7	39	53.9	2023-08	2025-02
KXCREDEFMAX	Commercial real estate high- est default rate	1	5	45.2	2023-08	2025-02
KXMORTGAGEDEF	Mortgage defaults	6	18	11.8	2023-08	2024-11
KXMANOFFVAC	Manhattan office vacancy rate	1	1	1.5	2026-08	2027-01

Notes: All series classified as real estate, sorted by volume within group. Titles as given by the exchange, truncated. Last close can lie in the future for open markets.

Table 7: **Kalshi volume by category**

Category	Series	Contracts (millions)	Share (%)
Sports	3,821.0	129,881	73.3
Crypto	275.0	35,087	19.8
Politics	2,404.0	2,412	1.4
Elections	1,872.0	2,009	1.1
Commodities	86.0	1,602	0.9
Mentions	453.0	1,263	0.7
Economics	835.0	1,204	0.7
Climate and Weather	415.0	1,181	0.7
Entertainment	2,547.0	1,174	0.7
Financials	989.0	1,028	0.6
Other categories	890	329	0.2
Real estate (within Economics)	65	3.45	0.002
House prices (within Economics)	28	0.35	0.000

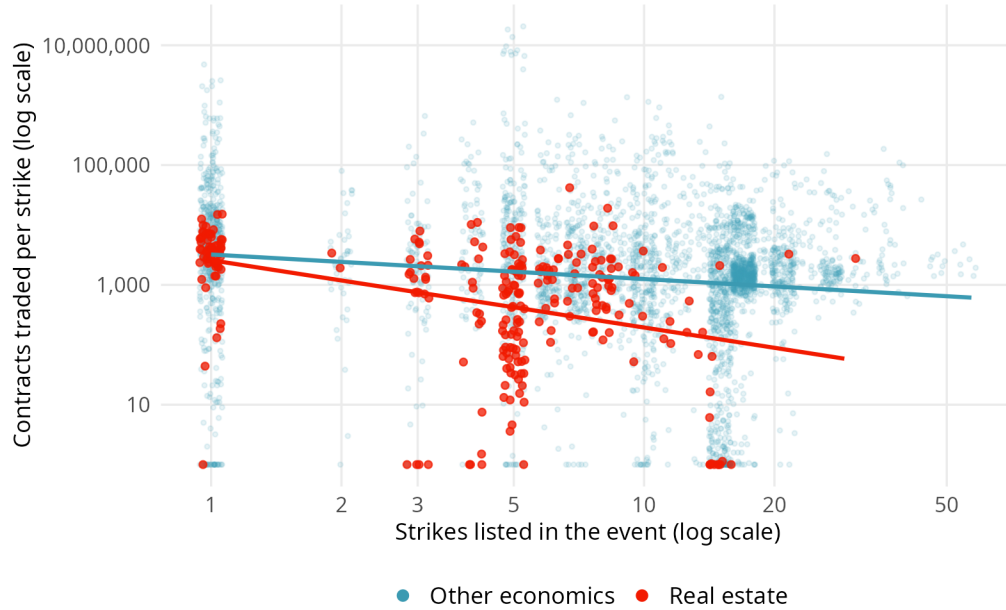
Notes: Lifetime volume of every series listed on Kalshi, by the exchange's own category, as of 5 October 2026. Categories with less than half a per cent of volume are grouped. The last two rows repeat the real estate series of this paper, which are part of the Economics category.

Table 8: **Summary statistics, closed events**

	N	Mean	SD	Min	Median	Max
Contracts traded	4,036	271,184.09	3,052,060.38	0.00	16,800.61	103,205,252.53
Contracts traded per strike	4,036	49,221.62	612,028.96	0.00	1,637.96	20,641,050.51
$\log(1 + \text{contracts traded})$	4,036	9.08	3.10	0.00	9.73	18.45
Strikes per event	4,036	11.26	7.79	1.00	10.50	57.00
Horizon (days)	4,036	36.96	82.15	0.04	6.58	1,006.89
Real estate event (0/1)	4,036	0.07	0.25	0.00	0.00	1.00
House price event (0/1)	4,036	0.01	0.08	0.00	0.00	1.00

Notes: One observation per event that closed before 5 October 2026. Contracts traded is summed over the strikes of the event. The last two rows are indicators: an event is a real estate event if its series belongs to any of the five real estate groups of Table 1, and a house price event if it belongs to the first of them, so their means are the shares of such events in the sample.

Figure 2: **Contracts traded per strike against the number of strikes**



Notes: Each point is one closed event. The horizontal axis is the number of strikes (markets) listed in the event, slightly jittered, and the vertical axis is the event's volume divided by that number, plotted as one plus the value so that events without any trade appear at the bottom. Lines are least-squares fits in logs for each group.

Table 9: **Median quoted spread by number of strikes in the event**

Strikes	Other economics		Real estate	
	Markets	Spread (cents)	Markets	Spread (cents)
1	110	4	60	3
2–3	79	5	58	4
4–6	497	5	361	6
7–10	1,164	6	380	6
11–15	1,157	6	134	8
16+	957	5	44	2

Notes: Settled markets with quote data, grouped by the number of markets listed in the same event. The spread is the median across markets of each market's median quoted spread.

Table 10: **Summary statistics, markets with quote data**

	N	Mean	SD	Min	Median	Max
Median quoted spread	5,001	0.121	0.204	0.010	0.060	0.980
Share of days with two-sided quote	5,999	0.625	0.389	0.000	0.786	1.000
Share of days with a trade	5,999	0.386	0.333	0.000	0.312	1.000
Price one day before close	3,997	0.487	0.375	0.010	0.485	0.990
Brier score, one day	3,997	0.098	0.179	0.000	0.009	0.980
Price seven days before close	2,560	0.461	0.359	0.010	0.435	0.990
Brier score, seven days	2,560	0.114	0.194	0.000	0.017	0.941
Horizon (days)	5,999	50.973	84.496	2.041	25.666	748.301
Real estate market (0/1)	5,999	0.207	0.405	0.000	0.000	1.000

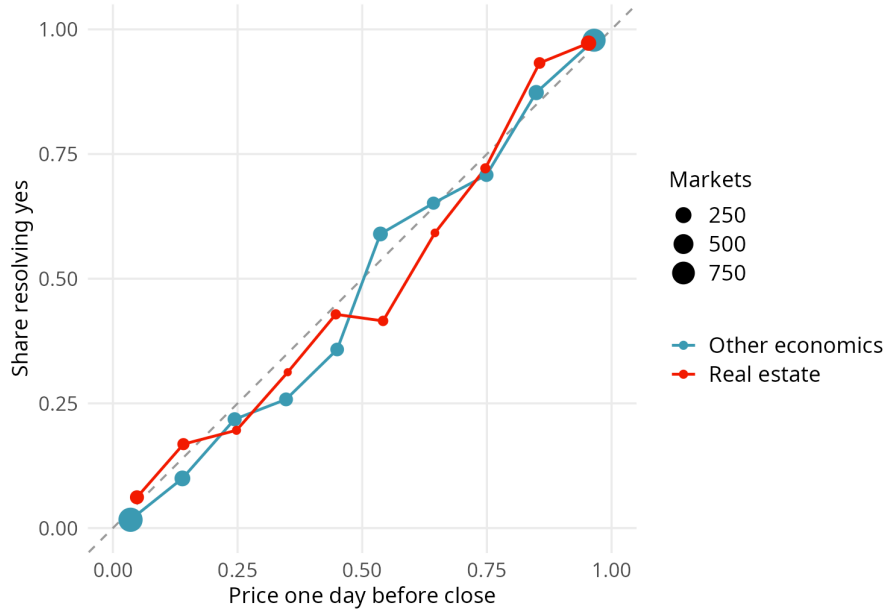
Notes: Settled markets listed for at least two days: all real estate markets and a fixed sample of other economics markets, chosen by a hash of the market identifier so that it stays the same across updates. Spreads are in dollars per one-dollar contract and are defined on days with both a bid and an ask. The Brier score is the squared difference between price and outcome. The last row is an indicator for markets in any of the five real estate groups.

Table 11: **Quotes and forecast accuracy by group**

	Share of days (%)				One day		Seven days	
	Markets	Two-sided	Traded	Spread (cents)	N	Brier	N	Brier
House prices	160	86	25	8	136	0.133	115	0.133
Rents	185	68	22	8	138	0.138	130	0.118
Mortgage rate	630	56	48	4	410	0.097	35	0.046
Housing activity	206	96	38	6	188	0.158	195	0.161
Credit and CRE	61	87	31	5	34	0.050	39	0.063
Other economics	4,757	61	39	6	3,091	0.091	2,046	0.110

Notes: Settled markets with quote data. Shares of days are means across markets, the spread is the median across markets of each market's median quoted spread, and N is the number of markets with a forecast at that lead.

Figure 3: Calibration one day before close



Notes: Markets are grouped into ten bins by price one day before close. Each point is the mean price and the share of markets in the bin that resolved yes.

Table 12: Share of house-level price variance removed by a Case-Shiller Chicago hedge

Holding (years)	SD of log change				R^2 of hedge		
	Pairs	House	Index	Hedge ratio	Linear	Seven binaries	One binary
0.5–1	25,608	0.440	0.045	-0.93	0.009	0.018	0.005
1–2	44,613	0.404	0.075	0.30	0.003	0.009	0.009
2–3	52,525	0.331	0.120	1.14	0.168	0.170	0.173
3–5	94,034	0.346	0.192	1.09	0.367	0.357	0.310
5–10	123,549	0.441	0.271	1.08	0.439	0.436	0.374
10+	126,073	0.383	0.231	0.93	0.314	0.306	0.151
All	466,402	0.403	0.211	0.95	0.248	0.260	0.209

Notes: Repeat-sale pairs of houses and condominiums in Cook County, Illinois, sold from 2000 on, warranty deeds. Each row regresses the log price change of the house on the hedge payoff over the same months. Linear is the log change of the S&P/Case-Shiller Chicago index; the hedge ratio is its slope. One binary pays if the index rose. Seven binaries have strikes at evenly spaced quantiles of the index change within the row.

Table 13: **Linear index hedge by purchase period, Cook County**

Holding (years)	Pairs	SD of index change	Hedge ratio	R^2
<i>Bought 2000–2011</i>				
0.5–1	14,737	0.051	-1.41	0.034
1–2	27,765	0.085	0.44	0.011
2–3	32,364	0.143	1.23	0.239
3–5	53,193	0.238	1.13	0.435
5–10	76,514	0.247	1.11	0.363
<i>Bought 2012 or later</i>				
0.5–1	10,871	0.035	0.09	0.000
1–2	16,848	0.053	-0.23	0.001
2–3	20,161	0.067	0.36	0.008
3–5	40,841	0.088	0.52	0.041
5–10	47,035	0.109	0.66	0.068

Notes: The linear hedge of Table 12, estimated separately by the date of the first sale in each pair.

Table 14: **Linear index hedge in alternative samples**

Holding (years)	Cook County, warranty deeds		Cook County, all deeds		New York City	
	Pairs	R^2	Pairs	R^2	Pairs	R^2
0.5–1	25,608	0.009	52,486	0.009	16,703	0.001
1–2	44,613	0.003	68,361	0.000	12,875	0.002
2–3	52,525	0.168	71,314	0.197	6,500	0.000
3–5	94,034	0.367	130,405	0.415	7,808	0.000
5–10	123,549	0.439	190,072	0.480	6,359	0.001
10+	126,073	0.314	162,214	0.321		
All	466,402	0.248	674,852	0.258	50,245	0.001

Notes: R^2 of the house's log price change on the log change of the S&P/Case-Shiller index for the city. The New York sales file starts in 2016, so no pair is held for ten years.

Table 15: **Polymarket house price events by launch month**

Launched	Events	Contracts (000)	Median	Paid (\$000)	Days	Wallets
January 2026	14	445	20,670	250	30	1,248
February 2026	8	144	18,194	78	33	425
March 2026	8	54	6,828	31	31	264
May 2026	8	45	5,554	34	27	154
June 2026	8	58	6,212	44	28	313
July 2026	8	779	100,315	397	102	1,396
October 2026	10	30	3,165	14	102	181

Notes: Events grouped by the month they opened. Contracts is volume as reported by the exchange and median is the median across events. Paid is what takers paid for those contracts according to the trade records, days is the median listing horizon, and wallets is the number of distinct addresses on either side of a trade. The latest cohorts are still open.

Table 16: The largest wallets in Polymarket house price markets

Rank	Dollars	Share (%)	Trades	Markets	Launch months	Maker (%)	Net / gross	Behaviour	Markets, all	Rewards, dollars
1	109,107	8.1	301	36	1	0	0.94	Takes, holds	874	1,232
2	60,607	4.5	1,002	43	1	29	0.21	Takes, trades out	2,069	20,896
3	55,159	4.1	1,344	134	5	34	0.47	Takes, trades out	4,946	34,724
4	48,049	3.6	3,598	214	5	69	0.47	Rests, flat	214	22
5	40,767	3.0	40	38	2	0	1.00	Takes, holds	625	55
6	37,429	2.8	144	31	1	9	0.79	Takes, holds	3,120	24,705
7	30,898	2.3	1,307	85	3	93	0.10	Rests, flat	41,019	638,297
8	25,974	1.9	3,459	263	5	71	0.33	Rests, flat	99,541	84,022
9	21,617	1.6	454	138	6	5	0.46	Takes, trades out	6,337	216
10	16,753	1.2	114	39	2	1	0.72	Takes, holds	282	732
11	14,955	1.1	84	26	2	1	0.85	Takes, holds	651	21
12	14,824	1.1	903	67	3	97	0.16	Rests, flat	94,599	599,917
13	14,043	1.0	1,584	107	4	30	0.38	Takes, trades out	107	0
14	13,859	1.0	613	140	4	15	1.00	Takes, holds	133,158	45,531
15	13,115	1.0	332	87	3	11	1.00	Takes, holds	41,183	2,454
16	12,177	0.9	312	47	1	5	0.35	Takes, trades out	4,020	1,066
17	10,679	0.8	625	90	3	12	1.00	Takes, holds	60,064	4,942
18	10,179	0.8	1,058	185	4	0	0.97	Takes, holds	73,124	0
19	10,096	0.8	388	51	2	99	0.02	Rests, flat	21,073	255,499
20	9,969	0.7	136	36	1	2	0.43	Takes, trades out	1,098	704
21	9,969	0.7	10	5	1	100	0.00	Rests, flat	787	128
22	9,345	0.7	247	16	2	30	0.00	Takes, trades out	11,193	103,493
23	9,025	0.7	536	110	3	6	1.00	Takes, holds	80,587	3,446
24	8,937	0.7	109	40	2	2	0.94	Takes, holds	68	0
25	8,850	0.7	568	103	6	54	0.37	Rests, flat	2,727	3,245

Notes: Wallets ranked by dollars traded in house price markets, both sides. Maker is the share of those dollars from resting orders. Net / gross is the absolute net position over gross trading, summed across markets. The last two columns cover the wallet's activity on the whole exchange: markets ever traded and liquidity rewards received, the latter from at most the 500 most recent payments. Addresses are not reported.

Table 17: **Price impact of orders in open house price contracts**

Order	Kalshi			Polymarket		
	Fillable (%)	Near (%)	Move (cents)	Fillable (%)	Near (%)	Move (cents)
100	99	99	0	99	75	1
1,000	78	29	16	39	5	60
7,000	19	4	51	1	0	84
50,000	0	0	–	0	0	–

Notes: Order books of open house price contracts on 5 October 2026, one observation per market and side. Order is the number of contracts bought. Fillable is the share of book sides holding at least that number at any price, and near the share that can fill the order without the price moving more than five cents from the best quote. Move is the median distance in cents from the best quote to the price of the last contract filled, among book sides that can fill the order. A hedge of 50,000 dollars takes 50,000 contracts at one strike or about 7,000 at each of seven.